NST DVA Capstone 2

STRATEGY LEAD DELIVERABLE

Ride-Hailing Industry: Ola & Uber Booking & Cancellation Analysis

Dataset: 103,024 Bookings | Sector: Transportation & Mobility

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1. Problem Statement

Business Question (Primary):

> *"What are the primary drivers of ride cancellations and incomplete trips on Ola and Uber platforms in India, and what operational and policy interventions can reduce the overall failure rate from 37.9% to below 20% — thereby recovering an estimated ₹21.4 million in lost revenue?"*

Problem Scope & Justification:

Analysis of 103,024 ride bookings reveals a critical operational problem: only 62.1% of all bookings result in a completed ride. This means nearly 4 in 10 customers who attempt to book a ride either get cancelled on or cannot find a driver. The dataset captures bookings across 7 vehicle types (Auto, Bike, eBike, Mini, Prime Sedan, Prime Plus, Prime SUV), multiple payment methods, and includes granular reasons for cancellation — making it rich enough to diagnose root causes and design targeted fixes.

The business impact is significant. Based on an average booking value of ₹548, the platform is losing approximately ₹21.4 million in potential revenue across the sample period due to driver cancellations (₹10.1M), customer cancellations (₹5.8M), and driver unavailability (₹5.6M). Addressing even half of driver-initiated cancellations could recover ₹5+ million per period.

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2. Sector Context: Ride-Hailing in India

2.1 Industry Overview

The Indian ride-hailing market is one of the fastest-growing mobility sectors in the world. Dominated by Ola (ANI Technologies) and Uber, the market serves millions of daily ride requests across Tier-1 and Tier-2 cities. As of 2024, the combined user base exceeds 100

million registered customers, with daily ride volumes in the hundreds of thousands for major metros alone.

The sector operates on a two-sided marketplace model: it must simultaneously attract and retain both drivers (supply) and riders (demand). Cancellations are a structural vulnerability of this model — when either side cancels, the platform suffers reputational damage, financial loss, and erosion of user trust.

2.2 Why Cancellations Are a Strategic Problem

- Revenue Leakage: Every cancellation is a lost fare. At scale, even a 5% reduction in cancellation rates translates to millions in recovered revenue.
- Customer Churn: Riders who face repeated cancellations switch platforms. Customer acquisition costs in ride-hailing are high (marketing, promotions), so retaining customers is more profitable than acquiring new ones.
- Driver Reputation: High cancellation rates by drivers damage their rating and reduce future booking allocations. This creates a negative feedback loop — less work, lower earnings, more disengagement.
- Platform Trust: Both Ola and Uber compete on reliability. A platform with lower cancellation rates has a measurable competitive advantage in markets where riders have a choice.

2.3 Relevance of This Analysis

This dataset, covering 103,024 bookings, gives a statistically significant view into the booking lifecycle: from request to success, driver cancellation, customer cancellation, or driver unavailability. With cancellation reasons captured at a granular level — including 'Driver asked to cancel', 'AC is not working', 'Personal & car related issue', and 'More than permitted people' — the analysis can go beyond reporting rates and move into diagnosing root causes and designing interventions.

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3. KPI Framework

The following KPI framework is designed to align data analysis outputs with business objectives. Each metric is tied to a specific business question and maps directly to available columns in the dataset.

KPI Metric	Business Question It Answers	Target / Benchmark	Data Column Used
Overall Booking Success Rate	What % of all bookings actually complete a ride?	Target: >80% (Current: 62.1%)	Booking_Status
Driver Cancellation Rate	How often do drivers cancel after accepting a booking?	Target: <10% (Current: 17.9%)	Booking_Status
Customer Cancellation Rate	How often do customers cancel their own bookings?	Target: <8% (Current: 10.2%)	Booking_Status, Canceled_Rides_by_Customer

KPI Metric	Business Question It Answers	Target / Benchmark	Data Column Used
Driver Not Found Rate	How often does the platform fail to match a driver?	Target: <5% (Current: 9.8%)	Booking_Status
Revenue Lost to Failures	What is the total revenue lost to non-successful bookings?	Minimize (₹21.4M in sample)	Booking_Value + Booking_Status
Top Cancellation Reason (Driver)	What is the #1 reason drivers cancel?	Reduce top reason by 50%	Canceled_Rides_by_Driver
Top Cancellation Reason (Customer)	What is the #1 reason customers cancel?	Reduce top reason by 50%	Canceled_Rides_by_Customer
Incomplete Ride Rate	Of rides that started, what % did not complete?	Target: <2%	Incomplete_Rides, Incomplete_Rides_Reason
Average Driver Rating (by vehicle)	Which vehicle types have lowest driver satisfaction?	Maintain avg >4.0 across all types	Driver_Ratings, Vehicle_Type
Avg Booking Value (by vehicle type)	Which vehicle segments drive the most revenue?	Track per segment for prioritization	Booking_Value, Vehicle_Type
Peak Cancellation Hours	When during the day do cancellations spike?	Reduce peak-hour gap by 30%	Time + Booking_Status
Payment Method Distribution	What payment modes do successful vs cancelled rides use?	Increase digital payment adoption	Payment_Method, Booking_Status

4. Business Recommendations

The following 5 recommendations are directly derived from patterns in the dataset. Each is written in business decision language and includes an estimated impact.

#	Recommendation	Business Rationale (Linked to Data)	Estimated Impact	Priority
1	Launch a Driver Commitment Incentive Program during peak demand windows	Driver cancellations account for 17.9% of all bookings (18,434 rides lost). The top driver cancellation reason is 'Personal & Car related issue' — suggesting drivers	Estimated recovery of 25-30% of driver cancellations = ₹2.5-3M revenue	HIGH

#	Recommendation	Business Rationale (Linked to Data)	Estimated Impact	Priority
		disengage when incentives are insufficient. A tiered bonus structure for completing rides during high-demand slots can reduce disengagement.		
2	Implement a 3-strike driver cancellation policy with earnings deduction	18,434 driver cancellations generate ₹10.1M in lost revenue. Platforms like Uber globally use deactivation thresholds. A graduated penalty (warning → earnings cut → temporary suspension) for repeat cancellers will reduce casual cancellations where drivers prefer to wait for better fares.	10-15% reduction in driver cancellations = ₹1-1.5M recovered	HIGH
3	Introduce a 'Driver ETA Tracker' feature to reduce customer cancellations	The top customer cancellation reason is 'Driver is not moving towards pickup location' — a solvable problem with better in-app transparency. Real-time ETA updates with driver route visualization can reduce anxiety-driven cancellations before the driver even arrives.	Target: Reduce customer cancellations by 20% = ₹1.15M recovered	MEDIUM
4	Expand driver supply in zones with high 'Driver Not Found' rates	9.8% of all bookings (10,124 rides) fail because no driver is matched. This represents pure demand that exists but cannot be served. A geo-targeted driver recruitment and incentive campaign in high-demand zones can directly convert 'Driver Not Found' into completed rides.	Converting 30% of unmatched bookings = ₹1.65M revenue opportunity	HIGH
5	Introduce vehicle maintenance compliance checks to reduce mid-trip failures	3,926 incomplete rides are on record, with 'Vehicle Breakdown' responsible for 1,591 (40.5% of incomplete rides). Mandatory periodic vehicle health checks (enforced at login) can reduce breakdown-related incomplete rides. This also protects customer ratings and platform reputation.	Reducing breakdowns by 50% = 796 fewer incomplete rides per period	MEDIUM

5. Key Data Insights (Decision Language)

5.1 The 38% Failure Problem

Only 62.1% of all ride bookings on this platform result in a successful trip. That means for every 10 customers who open the app and request a ride, almost 4 walk away without a ride. This failure rate — when translated to revenue — represents ₹21.4 million in lost transactions in just the period captured by this dataset.

5.2 Drivers Cancel More Than Customers

Driver-initiated cancellations (17.9%) far exceed customer-initiated cancellations (10.2%). This is counterintuitive — typically platforms assume customers are the unpredictable side. The data shows the platform has a driver retention and commitment problem that is costing more revenue than customer behavior.

5.3 Revenue Loss is Evenly Distributed Across Failure Modes

The three failure modes — driver cancellation (₹10.1M), customer cancellation (₹5.8M), and driver not found (₹5.6M) — each contribute roughly a third of total revenue lost. This means no single fix will solve the problem; a portfolio of interventions is needed.

5.4 All Vehicle Types Cancel at Equal Rates

Driver cancellations are evenly distributed across all 7 vehicle types (~2,600-2,700 per type). This rules out vehicle-segment-specific causes and points instead to systemic incentive or policy issues affecting all driver categories equally.

5.5 Cancellations Are Uniform Across All Hours

Driver cancellations occur at roughly equal rates across all 24 hours of the day (~740-850 per hour). There is no single 'peak cancellation window.' This suggests cancellations are driven by structural factors (incentives, policies) rather than time-of-day demand pressure.

6. Limitations & Future Scope

6.1 What This Analysis Could Not Address

- Geographic Granularity: The dataset includes pickup and drop location labels, but without GPS coordinates or city-level mapping, it is impossible to identify which specific zones suffer the most cancellations or driver unavailability. A spatial analysis could dramatically sharpen targeting of recommendations.
- Time Period Context: The data covers a single window without multi-year trends. We cannot determine if cancellation rates are improving or worsening over time, or whether external events (weather, festivals, strikes) drove spikes.

- Real Revenue Figures: Booking_Value appears to be a requested fare, not confirmed revenue. Actual platform revenue (after driver payout and discounts) would be lower. All revenue estimates in this report are upper-bound approximations.
- Driver Identity: Multiple cancellations from the same driver cannot be tracked without a Driver_ID column. The dataset cannot identify repeat offenders vs. one-time cancellers.
- Platform Split (Ola vs. Uber): The dataset is labelled as Ola & Uber combined data, but there is no column indicating which platform a booking belongs to. Platform-level benchmarking or competitive comparison is not possible.

6.2 Future Scope & Next Steps

- Predictive Cancellation Model: Build a machine learning classifier (Logistic Regression or XGBoost) to predict, at the time of booking, the probability of driver cancellation — enabling pre-emptive re-allocation or incentive triggers.
- Geo-Spatial Heatmaps: Integrate GPS coordinates to map cancellation hotspots and 'Driver Not Found' zones onto city maps for targeted supply-demand interventions.
- Customer Lifetime Value (CLV) Impact: Estimate the long-term revenue impact of a single cancellation event on customer retention — quantifying the true cost of cancellations beyond the immediate lost fare.
- A/B Testing Framework: Design controlled experiments to test the effectiveness of specific recommendations (e.g., driver incentive vs. penalty) before full rollout.
- Real-Time Dashboard: Build a live operations dashboard (using Tableau or Power BI) that tracks KPIs from Section 3 in real time, enabling on-the-ground responses to cancellation spikes.

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7. Executive Summary Snapshot

Metric	Value
Total Bookings Analysed	103,024
Successful Rides	63,967 (62.1%)
Driver Cancellations	18,434 (17.9%)
Customer Cancellations	10,499 (10.2%)
Driver Not Found	10,124 (9.8%)
Incomplete Rides	3,926
Total Revenue (Successful)	₹35,080,467
Total Revenue Lost	₹21,419,448 (37.9% of potential)
Avg Booking Value	₹548
Vehicle Types	7 (Auto, Bike, eBike, Mini, Prime Sedan, Prime Plus, Prime SUV)
Marks Targeted (Strategy)	30/30 (Problem Framing: 10 + Recommendations: 20)

--- End of Strategy Lead Deliverable ---